RAHUL RAWAT

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PROFESSIONAL SUMMARY

Data Scientist & AI Engineer with over 2 years of hands-on experience delivering end-to-end ML and AI projects using deep learning, NLP, and AWS. Skilled at building scalable models and intelligent systems that turn data into real-world impact.

HANDS-ON EXPERIENCE

Team Leader (Capstone Project) **Jan 2024 – Aug 2025**
Lambton College *Mississauga, ON*

- Designed a FinTech AI system to extract insights from PDF financial reports, building a **LangChain + FAISS**-based RAG pipeline powered by **LLMs** for real-time document summarization and intelligent query answering.
- Fine-tuned a **BERT classifier** for span vs. arithmetic query detection, achieving **96.59% accuracy**, and deployed it via **AWS SageMaker** to reduce inference time from **2 minutes to 2 seconds**.

GitHub: FinChatbot

Data Science Intern **Jul 2021 – Aug 2021**
Diginique TechLabs *Delhi, India*

- Developed and fine-tuned multiple **regression models** to predict continuous outcomes, achieving an average R^2 **score of 0.90**, outperforming baseline benchmarks by 5%.
- Enhanced model robustness by optimizing feature selection, automating preprocessing pipelines, and integrating results into a scalable evaluation framework.

PROJECTS

RAG Pipeline – AI-Powered Document Analysis and Chat System (Project Link):

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** system using **LangGraph**, **FastAPI**, and **Streamlit**, enabling users to upload multiple PDF and TXT documents and ask natural-language questions grounded in their content.
- Developed a multi-stage **LangGraph workflow** with specialized agents for query rewriting, document retrieval, relevance grading, answer generation, and fallback handling to improve retrieval relevance and reduce unsupported responses.
- Implemented session-specific **FAISS vector indexes** using **OpenAI text-embedding-3-small** embeddings, with asynchronous **MongoDB** integration for document metadata storage and conversation checkpointing.
- Built asynchronous **FastAPI** endpoints for document uploads and querying, integrated with a **Streamlit chat interface** featuring multi-file uploads and session-based conversation history; containerized the application using **Docker**.

CaptionAI – Image Captioning using Encoder-Attention-Decoder Architecture (Project Link):

- Designed a deep learning-based **image captioning system** using a modular **Encoder–Attention–Decoder** architecture in PyTorch, trained on the **Flickr8k dataset**.
- Used a **pre-trained ResNet-50 CNN** as the encoder to extract visual features, followed by an **LSTM decoder** with an **attention mechanism** to generate context-aware captions.
- **Implemented modular code design** separating feature extraction, sequence generation, and attention logic for improved readability, debugging, and scalability.

- Integrated with a **Streamlit-based UI** allowing users to upload images and receive generated captions in real time; future plans include BLEU score evaluation and transformer-based upgrades.

SKILLS

Data Science & AI: Python (Pandas, NumPy, Scikit-learn), **Deep Learning** (PyTorch, TensorFlow), **LLMs**, **NLP**, **Embeddings**, **RAG**, **SQL/MySQL**, **MongoDB**.

Frameworks & Tools: LangGraph, **LangChain**, **FastAPI**, **Docker**, **Git**, **Power BI**.

Platforms & Infra: AWS (S3, EC2, SageMaker), **Vector Databases** (FAISS, Chroma, Pinecone), **Streamlit**, **Jupyter Notebook**.

EDUCATION

Postgraduate Certificate in Big Data Analytics, **Lambton College** **Jan 2024 – Aug 2025**

Coursework: Big Data Tools, Neural Networks and Deep Learning, Big Data Algorithms and Statistics, Big Data Visualization, Natural Language Processing and Social Media Analytics, NoSQL (MongoDB, Neo4j, Redis).

College Projects:

- **Pneumonia Detection CNN** (Project Link):
  - Built a multi-model pipeline (Logistic Regression, Random Forest, CNN, and VGG16 with Transfer Learning) to classify X-ray images, achieving **91.67% accuracy**.
  - Deployed with a **Streamlit UI** and modular codebase.
- **Sentiment Analysis using NLP** (Project Link):
  - Preprocessed noisy text data using **regex** and vectorized using **BoW and TF-IDF**.
  - Compared **3 ML models** on binary sentiment classification, with thorough evaluation of performance metrics.

Bachelor of Science in Mathematics, **Panjab University** **Jul 2019 – Aug 2022**

Achieved 77% overall with 85% in mathematics coursework.

Relevant Subjects: Linear Algebra, Calculus I & II, Probability Theory, Numerical Analysis, Differential Equations.

Applied Work:

- Built regression and classification models **from scratch** using concepts from **Linear Algebra and Calculus**, implemented entirely in **NumPy**.

CERTIFICATIONS

- Machine Learning Specialization: Verify the certification here.
- TensorFlow Developer: Verify the certification here.